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Title: Automated Structural Health Monitoring: Facade and Concrete Crack Detection Using YOLOv11

1. Project Description

Problem Statement:

Ensuring the preservation of aging infrastructure constitutes a significant challenge to urban safety. Manual inspections of building facades and concrete structures are labor-intensive, costly, and prone to human error, particularly when assessing extensive complexes or high-rise edifices. Small cracks often serve as preliminary indicators of potential structural failure; however, they are frequently neglected during routine visual evaluations.

Project Objective:

This project develops a comprehensive computer vision pipeline aimed at automating the detection of structural cracks. Employing the YOLOv11 (You Only Look Once) architecture, the model is trained to accurately identify and localize cracks in real-time, thereby equipping inspectors with a digital tool to effectively prioritize high-risk areas.

Technical Approach:

- Architecture: YOLOv8n (Nano) for high-speed inference in cloud and mobile environments.
- The dataset consists of an extensive compilation of approximately 4,000 annotated images, illustrating a diverse range of concrete surfaces, lighting conditions, and crack geometries.
- Platform: Cloud-based training emphasizing reproducibility via keyless data integration.

Impact & Significance:

Automating this process facilitates more frequent inspections at a decreased expense. Transitioning from "reactive" maintenance to "proactive" monitoring, this technology assists in extending the lifespan of infrastructure and substantially reduces the risk of catastrophic structural failures.

2. Diversity & Inclusion Statement

This project is dedicated to an inclusive approach to urban safety. By offering an open-source framework for infrastructure monitoring, we seek to reduce the barriers faced by municipalities in under-resourced areas in accessing advanced safety technologies. Our dataset selection intentionally encompasses older structures and various masonry styles to guarantee that safety innovations serve all communities, irrespective of the age or prominence of their infrastructure.